

MOHAMMAD JAMIL SHUVO

University of Freiburg, Freiburg, Germany



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RESEARCH INTERESTS

Ecological geneticist studying how environmental heterogeneity shapes biodiversity, population structure, and adaptive responses across ecosystems. I integrate eDNA-based approaches, community modelling, and landscape genomics to link species composition, genetic variation, and ecological interactions.

KEY COMPETENCIES

Biodiversity monitoring · arthropod community ecology · eDNA sampling, metabarcoding and metagenomics · molecular laboratory work · large biodiversity datasets · QA/QC and workflow automation · ecological modelling · landscape and conservation genomics · bioinformatics with R, Python, Bash, C++ and HPC · scientific writing and collaboration.

EDUCATION

01/07/2022 – current Freiburg, Germany

PhD Candidate - University of Freiburg - **Dissertation submitted**; disputation scheduled for 6 November 2026

Project: “Genetic assessment of saproxylic arthropods across the Black Forest” (DFG RTG CONFOBI B4)

30/09/2019 – 29/04/2022 Gießen, Germany

Master of Science (MSc) in Agrobiotechnology - Justus Liebig University (JLU)

31/05/2014 – 09/10/2018 Khulna, Bangladesh

Bachelor of Science (BSc) in Biotechnology and Genetic Engineering - Khulna University

RESEARCH & PROFESSIONAL EXPERIENCE

15/11/2026 – 30/11/2026 (Planned) Cambridge, UK

Visiting Research Fellow - Insect Evolution and Genomics Group, University of Cambridge

PoMito: A reproducible workflow for mitochondrial genome assembly, annotation, and phylogenomic analysis from individual and pooled low-coverage short-read data.

01/07/2022 – Current Freiburg, Germany

Scientific Employee - Chair of wildlife ecology and management, University of Freiburg

Forest arthropod biodiversity, eDNA metabarcoding, landscape genomics and reproducible workflows across 135 managed forest plots and 82 *Entomobrya nivalis* populations.

- Developed eDNA and Pool-seq workflows for biodiversity monitoring and non-model invertebrate genomics.
- Analysed biodiversity, environmental and genomic datasets using R, Bash, Python and HPC.
- Linked forest structure, land use and environmental gradients with community and genomic responses.

31/01/2021 – 29/04/2022 Frankfurt am Main, Germany

Guest Researcher - Senckenberg Society for Nature Research (SGN), SBIK-F

Molecular biodiversity assessment of soil invertebrates within the MetalInvert project, with a focus on Collembola mitochondrial genome assembly, annotation and biodiversity monitoring.

- Developed Python/Bash pipelines for mitogenome assembly, annotation and phylogenetic analysis.
- Created a Collembola mitogenome resource with 120 annotated mitochondrial sequences.
- Evaluated taxonomic resolution and ecological patterns using metagenomic and mitochondrial datasets.

31/05/2017 – 30/01/2019 Khulna, Bangladesh

Research Assistant - Plant biotechnology laboratory, BGE discipline, Khulna University

Worked on molecular identification of fungal and bacterial isolates and evaluation of salt-tolerant maize hybrids.

PROFESSIONAL DEVELOPMENT

- Six-month-long “[Google Data Analytics](#)” online professional course (December 18, 2024)
- **Adaptation Genomics, [Physalia Courses \(online\)](#)**: 8-12 December 2025 - Training in population and comparative genomics methods for studying adaptation from whole-genome data.

- **Working Group Member**, COST Action CA22129: **InsectAI** (European Cooperation in Science and Technology), 2026 - Present.

FUNDING & AWARDS

- DFG RTG ConFoBi PhD project - (65%, TV-L E 13) – including six months of parental leave
- DAAD (Deutscher Akademischer Austauschdienst) Scholarship (2020 & 2021) - €15 k total
- Presidential Commission Scholarship, JLU Giessen (2021)
- Khulna University merit scholarship (2014 – 2018)

OUTREACH & SUPERVISION

Active participation in RTG summer schools, forester meetings, retreats, workshops and symposia.

- Poster presentation, 6th European Conservation Genetics Meeting (August 2024, Lausanne): Uncovering the Hidden World of Deadwood Arthropods
- Participation in the Biodiversity Genomics Conference (2024, online)
- Oral presentation, Biodiversity Exploratories Conference (June 2026, Berlin): From Draft Genome to Landscape Genomics: A Reproducible Pool-seq Framework Reveals High Connectivity and Weak Polygenic Adaptation in a Soil Springtail
- Poster presentation, 7th European Conservation Genetics Meeting (August 2026, Antwerp): High connectivity and subtle genotype-environment associations in the soil springtail *Entomobrya nivalis* across a managed forest landscape
- Co-supervised an MSc thesis and an internship project

TEACHING

Lecturer, WiSe 2024 & 2025 - Conservation of Forest Biodiversity course (MSc module) - University of Freiburg

Delivered practical training from eDNA sampling to data analysis for second-year MSc students, contributing to follow-up thesis and internship projects.

Creator & instructor, online “Data Analysis with R” (2023 – present; 100+ participants) - Khulna University, Bangladesh

Course covered essential topics of ecological data analysis such as data manipulation with the tidyverse package, data visualization with ggplot2, hypothesis building, and statistical modeling.

SELECTED PUBLICATIONS

1. **Shuvo, M. J., et al.** High connectivity and subtle genotype-environment associations in *Entomobrya nivalis*. Manuscript under internal review.
2. **Shuvo, M. J.** PoMito: Recovering mitochondrial genomes from pooled and individual low-coverage short-read data for ecological and evolutionary genomics. Manuscript in preparation.
3. **Shuvo, M. J., et al. (2026)**. PoLoCo: A reproducible pooled low-coverage workflow for non-model invertebrates. *GigaByte*. Under review. **Workflow DOI:** <https://doi.org/10.5281/zenodo.19601574>
4. **Shuvo, M. J., et al. (2026)**. Influence of deadwood, tree-related microhabitats, and forest structural features on saproxylic arthropod diversity. *Ecology and Evolution*, 16, e73600. <https://doi.org/10.1002/ece3.73600>
5. **Shuvo, M. J., et al. (2025)**. Optimizing eDNA metabarcoding techniques for assessing arthropod communities in tree-related microhabitats. *Environmental DNA*, 7, e70064. <https://doi.org/10.1002/edn3.70064>
6. Ahmed, S., et al. (2022). Salinity reduces site quality and mangrove forest functions. *Science of the Total Environment*, 853, 158662. <https://doi.org/10.1016/j.scitotenv.2022.158662>

LANGUAGE SKILLS

Bengali (native), English (C1), German (B1) & Hindi (A2)

LEADERSHIP

Elected Member of the Executive Committee (2025–2027), Khulna University BGE Alumni Association (KUBAA)

Executive editor of KUBAA YouTube and monthly outreach bulletin

Founding member of HELIX (Biotech Career Development Club)

Founding vice-president and constitution writer of Mahilara College Alumni Association (MCAA)

REFEREES

Prof. Dr. Gernot Segelbacher

PhD supervisor

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Prof. Dr. Katrin Heer

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